SMARTPHONE SENSING • AFFECTIVE COMPUTING • MOBILE MENTAL HEALTH

MoodCam

Can your phone's front camera quietly learn your mood — just from the selfies it snaps every time you unlock it?

25

Real people, 4 weeks

15,995

Real-world phone snapshots

0.64

Best AUC score

Presented from: Islam, Zhang & Bae — "MoodCam" (arXiv:2412.12625)

How MoodCam Works

Every time you unlock your phone or open an app, the front camera briefly sees your face anyway (think Face ID). MoodCam piggybacks on that moment — no extra effort from you.

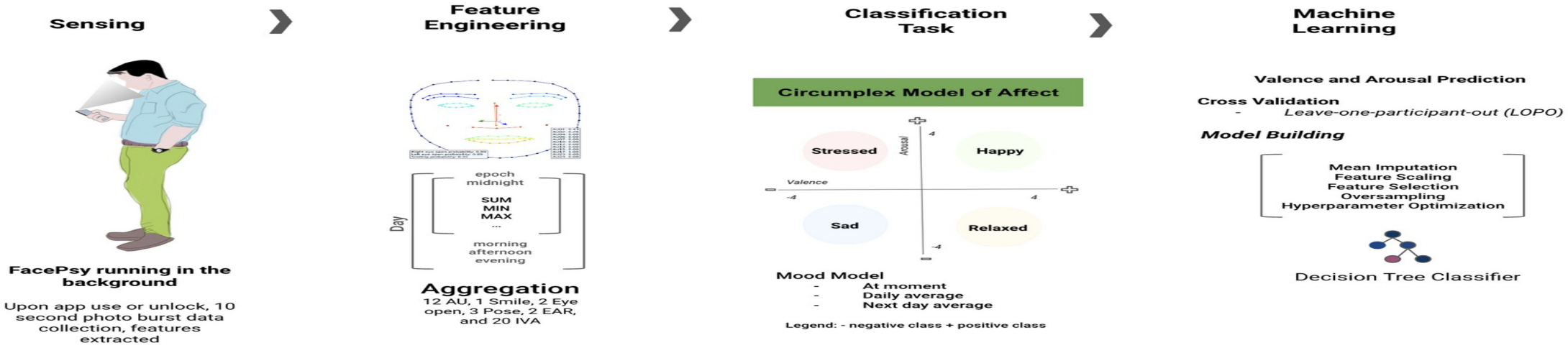


Figure 1: the four-step pipeline — Sense a photo burst → Engineer facial features → Classify into the "Circumplex" mood map → Train a machine-learning model.

1. Sense

An app called FacePsy runs quietly in the background and grabs a short photo burst whenever the phone is unlocked.

2. Two Mood Dimensions

Mood is scored on Valence (unpleasant ↔ pleasant) and Arousal (sleepy ↔ alert) — the well-known "Circumplex Model of Affect."

3. Predict

A Decision Tree model learns to classify each moment as high/low Valence and high/low Arousal from facial signals.

Collecting Real-World Data (Not a Lab)

Most past mood-detection studies used posed, lab-recorded, or online video datasets. MoodCam instead followed real people living their normal lives.

The Study, In Numbers

25 participants completed the 4-week study (started with 38)

15,995 facial data snapshots collected automatically

544 days of usable facial data across everyone

3x / day participants also self-reported mood via quick surveys (10am, 3pm, 8pm)

What the Camera Actually Measures

12 Action Units: small muscle movements — e.g. brow raise, lip corner pull — the building blocks of any facial expression.

Smile & Eye-Open Probability: how likely you are smiling, and how open your eyes are.

Head Pose (yaw / pitch / roll): which way your head is tilted or turned.

Eye Aspect Ratio: a simple measure of how "open" each eye looks.

Inter-Vector Angles: geometric angles between facial landmarks that describe overall face shape.

Three Ways to Predict Mood

The team built models that answer three different questions — each useful for a different kind of mental-health support.

At the Moment

"How do I feel right now?" (uses the last 30 minutes of facial data)

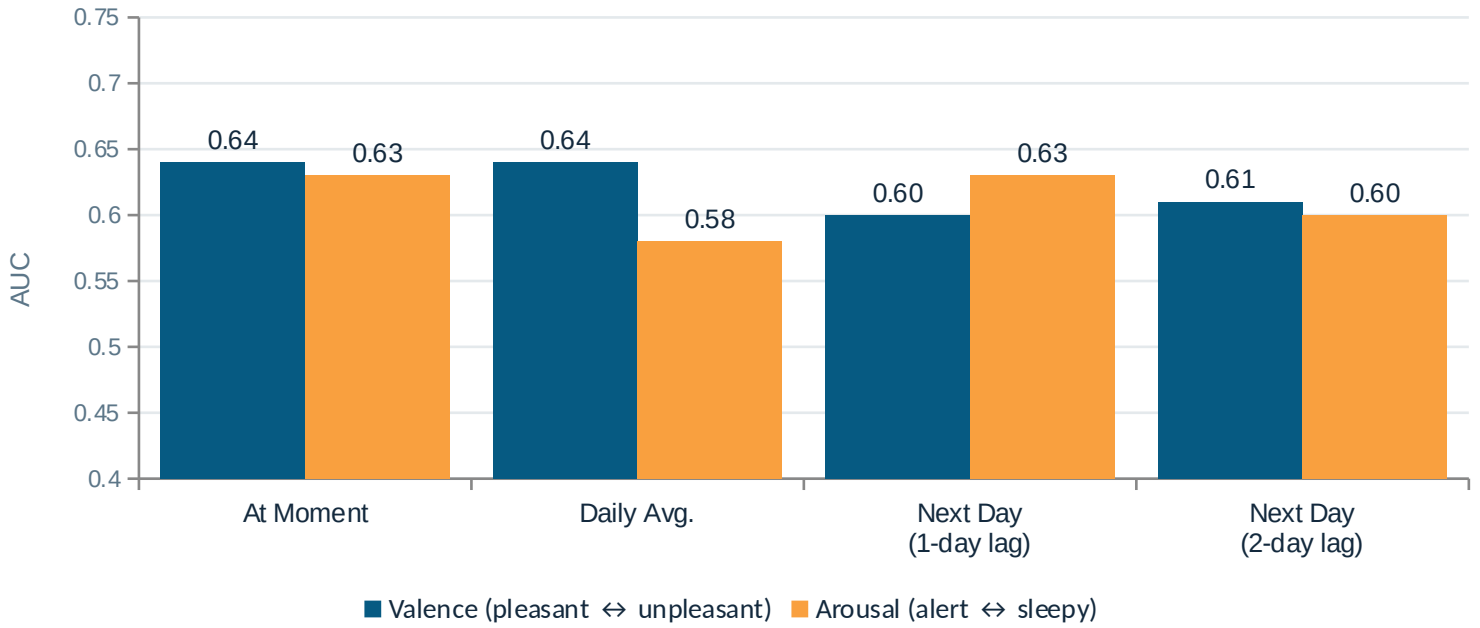
Daily Average

"What was my overall mood today?" (useful for spotting an off day)

Next-Day Forecast

"How will I likely feel tomorrow?" (helps plan support in advance)

Accuracy (AUC) for each model — 0.5 = random guess, 1.0 = perfect



Reading the chart

- All AUC scores land around 0.58–0.64 — clearly better than a coin flip, but not yet clinical-grade.
- Longer time windows (Daily, Next-Day) sometimes predict Valence even better than "in the moment."
- Performance stopped improving after a 2-day lag — older data adds little extra signal.

What We Learned

1 Eyes say a lot

The Eye Aspect Ratio (how open your eyes look) was consistently one of the strongest signals for mood — across almost every model.

2 Action Units matter most ‘in the moment’

Removing facial muscle-movement features hurt the “At Moment” Arousal model the most — they capture fast, fleeting emotional reactions.

3 Competitive with (and sometimes better than) prior work

MoodCam's AUC beat a prior mobile-sensing study (0.51) and its F1 scores beat another (0.53), though one touch/motion-based study still scored higher (0.90).

4 Real life is messier than a lab — and that's the point

Unlike posed lab videos, these are genuine unlocking-your-phone faces, making the (modest) results more believable for everyday use.

Why it matters: a phone you already unlock 80+ times a day could passively flag mood dips — no extra surveys, no wearables, just the camera you already have.